

Emiliano de la Garza Villarreal

Languages: English and Spanish

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EDUCATION

(B.S.) Data Science - University of Tennessee Chattanooga

Aug 2024 – May 2026

(A.S) Computer Science - Cleveland State Community College

Dec 2023

WORK EXPERIENCE

Machine Learning Researcher, CUIP

Apr 2025 – Present

- Built a real-time intersection monitoring system on NVIDIA Jetson using ROS2 and CUDA to deploy a Federated-trained model, implementing multi-processing to resolve CPU image-processing bottlenecks and ensure continuous GPU saturation for reliable, low latency, real-time inference.
- Achieved state-of-the-art (SOTA) trajectory prediction by designing a novel federated learning strategy within a Flowchain model, combining FedProx and FedAdam to improve ADE and FDE metrics ~8% compared to baseline results.
- Developed a custom YOLO crash detection model by gathering open-source data and applying transfer learning to existing weights, then iteratively improved accuracy and recall by experimenting with data augmentation techniques like Mosaic and MixUp to expand system capabilities.

Engineering Intern, Eaton Corporation

Dec 2023 – Present

- Analyzed large-scale industrial datasets to identify operational trends and inefficiencies, translating raw metrics into actionable insights to generate over \$300,000 in annual savings.
- Conducted shop-floor time studies to build a predictive production model, updating system standards that reduced labor hours by 20%.

TECHNICAL SKILLS

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- **Machine Learning & Deep Learning:** Proficient in developing and optimizing models with PyTorch, specializing in distributed systems, Computer Vision (YOLO), and real-time inference using NVIDIA TensorRT.
 - **Data Engineering & Processing:** Experienced in building real-time data pipelines for video stream processing (RTSP) and time-series analysis using Pandas, NumPy, and OpenCV.
 - **Programming & Development Tools:** Advanced proficiency in Python (including multiprocessing & threading), Elementary proficiency in C++ and Java, general Data Structures and algorithms, with sufficient knowledge in version control (Git), and decent skills with Linux system terminal.